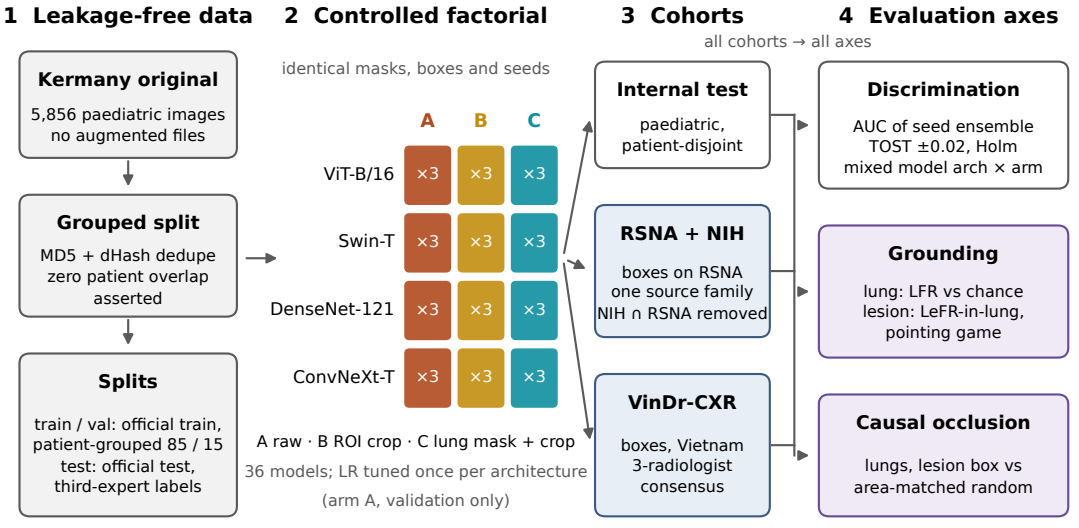
1 Leakage-free data
2 Controlled factorial
3 Cohorts
4 Evaluation axes
all cohorts → all axes
Kermany original
5,856 paediatric images
no augmented files
Grouped split
MD5 + dHash dedupe
zero patient overlap
asserted
Splits
train / val: official train,
patient-grouped 85 / 15
test: official test,
third-expert labels
identical masks, boxes and seeds
A
B
C
ViT-B/16
Swin-T
DenseNet-121
ConvNeXt-T
×3
A raw · B ROI crop · C lung mask + crop
36 models; LR tuned once per architecture
(arm A, validation only)
Internal test
paediatric,
patient-disjoint
RSNA + NIH
boxes on RSNA
one source family
NIH ∩ RSNA removed
VinDr-CXR
boxes, Vietnam
3-radiologist
consensus
Discrimination
AUC of seed ensemble
TOST ±0.02, Holm
mixed model arch × arm
Grounding
lung: LFR vs chance
lesion: LeFR-in-lung,
pointing game
Causal occlusion
lungs, lesion box vs
area-matched random